

Shared Fun Is Doubled Fun: Player Enjoyment as a Function of Social Setting

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Abstract. Although the social relevance of digital gaming has been discussed and investigated in cultural and sociological readings, social context has been largely neglected in terms of in-game player experience. In this paper we present a study in which player experience is empirically tested comprehensively, with the focus on the interplay between player enjoyment and social context. Self reports on the Game Experience Questionnaire and an aggression state questionnaire were explored for three play configurations: virtual, mediated, and co-located co-play. Additionally, effects of familiarity and winning vs. losing were tested. Results indicate that a co-located co-player significantly adds to the fun, challenge, and perceived competence in the game. Aggression was influenced by familiarity and performance. The effect of social context was mediated by the level of social presence. These findings illustrate that social context is an important determinant of player enjoyment and should be incorporated in models of player experience.

Keywords: Digital gaming, game experience, player experience, enjoyment, social presence, play setting, social setting, play configuration.

1 Introduction

Play is an intrinsically motivated, physical or mental leisure activity that is undertaken only for enjoyment or amusement and has no other objective [1]. Games present more structured and rule-based play modes, and are often more goal-directed. But both play and games are engaged in by people of all ages. In fact most people have played board games or card games with their parents, siblings, children, or friends. These traditional games are often seen as a constructive means for family/peer interaction.

As do traditional games, digital games offer opportunities for social interaction [2,3,4,5]. In fact, according to gamers, socializing is the number one motivation for playing digital games [3]. Yet besides playing with real and present friends, digital games also offer possibilities for playing against or with non-human co-players (i.e. computer agents) or mediated human co-players (i.e. online play). To date, little attention has been given to the nature and role of the exact social setting in explanations of player enjoyment. Apart from two psychophysiology-based studies on the effects of competing against a computer or a human [6,7] most studies concerning

player experience focus only on one specific [4] or even a non-specified [8,9] social play configuration. They have left unaddressed whether enjoyment is equal in collocated multi-play versus other popular play settings that do not include a human co-player (i.e., computer agent as co-player) or lack the physical nearness of a human co-player (i.e., online play).

In spite of the clear social roles digital games play, the social component of gaming is often missing in theories and models of player experience and digital gaming platforms are not valued to their full extent as a medium which brings people together [e.g., 8,9,10]. However, some researchers in the field have argued that player enjoyment will differ between settings that differ in potential for social interaction [11]. Physical play settings are a complex mix of social (with friends vs. with strangers), spatial (at home vs. at the Internet café) and media (side-by-side vs. online) characteristics. It is hypothesized that the impact of socio-spatial setting on player experience is substantial, and mediated by the social presence of the co-player [11]. Social presence is defined as the subjective counterpart of objective closeness in communication: 'the sense of being with another' [12] and is mainly applied in respect to communication technology. Recently, the concept was also introduced in digital gaming research [5].

The current study aimed to empirically test the influence of a co-player presence on player enjoyment, by systematically varying social play settings with increasing levels of objective co-player presence or closeness. The mediating role of social presence, the subjective sense of being with another, was also explored. Moreover, player enjoyment was treated and measured as a multidimensional construct, giving detailed insight into player experience and how it is shaped by situational characteristics.

1.1 Digital Game Enjoyment

Player experience is a broad term which can be roughly subdivided into player involvement and player enjoyment. Player involvement describes a player's focus and interest during digital play in terms of flow, immersion and engagement [13]. We regard digital game enjoyment as a generic term that indicates the amount of pleasure or displeasure and includes concepts such as positive affect, competence, challenge, frustration and aggression. Explicitly treating player enjoyment as a multidimensional construct, will give a better understanding of how player experience is shaped; for example a digital game can induce frustration and aggression but still can be experienced as a positive and challenging activity.

Because the digital gaming field has been lacking a common vocabulary, player experience is studied in a fragmented way [14], which gives an incomplete and imprecise picture of what player experience actually is. IJsselsteijn, de Kort, & Poels [15] therefore developed the Game Experience Questionnaire (GEQ) which enables game researchers to empirically explore and test player experience of digital games in a quantitative and comprehensive way. The questionnaire was based on focus group investigations [14] and two subsequent large scale survey investigations among frequent gamers [15].

The GEQ includes player enjoyment-related scales of positive affect, competence, challenge and frustration (besides more involvement-oriented scales probing flow, immersion, and boredom). Positive affect basically probes the fun and enjoyment of